

MOTOR STARTERS

Direct-on-line and star-delta control for
agricultural and industrial applications.

D.O.L. STARTERS

0.5 to 7.5 HP

STAR-DELTA STARTERS

1 to 15 HP



THREE ENCLOSURE FORMATS / ONE KVP IDENTITY

MARKETED BY

K V P TRADING COMPANY

H.No. 20, Flat No. 2, Temple Rock Enclave, Tadbund,
Secunderabad-9 | 93462 93463

PRODUCT OVERVIEW

Built for dependable starts

Motor starter range and operation from the original catalog.

The source GK-1 Direct-On-Line and Star-Delta starters have robust construction and offer smooth, reliable performance. They are designed for demanding agricultural and industrial fields.

01 **D.O.L. STARTERS**

RANGE / 0.5 TO 7.5 HP

Direct-On-Line starters consist of one GK-1 relay and one GK-1 contactor.

The contactor protects the motor against under voltage; the relay protects the motor against overloads.

02 **SEMI-AUTOMATIC STAR-DELTA**

RANGE / 1 TO 15 HP

Star, delta and line contactors are provided with an overload relay.

The motor draws about one-third of its normal starting current, so other loads on the same line are less affected.

03 **STAR-DELTA CHANGEOVER**

Semi-automatic: hold START until the motor reaches 80-85% speed in star; releasing START switches from star to delta.

Fully automatic: a precise thermal timer makes the changeover automatically in 10-15 seconds (source catalog figure).

TECHNICAL REFERENCE

D.O.L. rating table

Motor HP, relay range, and maximum / minimum back-up fuse ratings.

TABLE OF RATINGS FOR GK-1 D.O.L. STARTER

Motor Rating H.P.			RELAY RANGE Amps	Back - up Fuse Rating			
220 V 1 - Phase	400 V 3 - Phase	440 V 3 - Phase		Maximum		Minimum	
				Diazed	SWG	Diazed	SWG
	0.05 0.1 0.125 0.15	0.05 0.1 0.125 0.15	0.15-0.25 0.25-0.4 0.25-0.4 0.4-6.65	1A 2A 2A 4A			
	0.2 0.25 0.5	0.2 0.25 0.5	0.4-1.65 0.4-0.65 0.6-1.0 1.0-1.6	4A 4A 6A 6A	35 35 32 32		
0.125						6A 6A	32 32
0.25	0.75 1.0 1.25	0.75 1.0 1.25	1.0-1.6 1.5-2.5 1.5-2.5 2.5-4.0	6A 15A 15A 25A	32 25 25 23	6A 6A 6A 10A	32 32 32 28
0.75	1.5 1.75 2.0 2.5	1.75 2.0 2.5	2.5-4.0 2.5-4.0 2.5-4.0 4.0-6.5	25A 25A 25A 25A	23 23 23 23	10A 10A 10A 10A	28 28 28 28
0 1.0 1.25	3.0 4.0 5.0	3.0 4.0 5.0	4.0-6.5 4.0-6.5 6-10 6-10	25A 25A 35A 35A	23 23 21 21	10A 10A 15A 15A	28 28 25 25
1.5 1.75 2.0 2.0	6.0 7.5 7.5	6.0 7.5 7.5	6-10 9-14 9-14 11-18	35A 35A 35A 35A	21 21 21 21	15A 15A 15A 15A	25 25 25 25
2.0 3.0 3.0	10 12.5 15	10 12.5 15	13-21 20-32 20-32	35A 45A 45A	21 19 19	15A 25A 25A	20 20 20

Technical ratings reproduced from the supplied Ganga catalog for reference.
Confirm the final KVP model specifications before printing or production.

TECHNICAL REFERENCE

Star-delta rating table

400/440 V, three phase: motor current, relay and fuse selection.

TABLE OF RATINGS FOR GK-1 STAR-DELTA STARTER

Motor Rating 400/440V 3-Phase, H.P.	Full load line current in Amps I_n	$I_n/\sqrt{3}$	Relay Range Amps	Back - up Fuse Rating			
				Maximum		Minimum	
				Diazed	SWG	Diazed	SWG
1	1.7	1	1 - 16.A	6	32	4	35
1.5	2.4	1.4	1 - 1.6 A	15	25	6	32
2	3.1	1.8	1.5 - 2.5 A	15	25	6	32
3	4.4	2.5	2.5 - 4 A	25	23	6	32
4	5.7	3.3	2.5 - 4 A	25	23	10	28
5	7.1	4.1	4 - 6.5 A	25	23	10	28
6	8.4	4.8	4 - 6.5 A	25	23	10	28
7.5	10.4	6.0	2.5 - 4.A	35	21	15	25
10	13.6	7.9	6 - 10A	35	21	15	25
12.5	17	9.8	9 - 14A	35	21	20	24
15	19.9	11.5	13 - 21A	35	21	25	23
20	28	16.5	13 - 21A	35	21	25	23
20	28	20	20 - 32A	45	19	30	20

Technical ratings reproduced from the supplied Ganga catalog for reference.
Confirm the final KVP model specifications before printing or production.

SOURCE CATALOG OPERATING NOTES

Star-delta starters use star, delta and line contactors with overload relay.

The motor draws approximately one-third of normal starting current.

Semi-automatic changeover follows release of START at 80-85% speed.

Fully automatic switching uses a 10-15 second thermal timer.

COMPONENT DATA

Air-break contactor

Electrical and mechanical data reproduced at readable scale.

TECHNICAL DATA	
No. Of Main Poles	4
Nominal rating at 500V, 50 Hz	16A
Motor Duty: 30 415V, 50 Hz AC3 Duty	7.5 hp / 5.5 kw
Terminal capacity: wires upto	6mm ²
Auxiliary Contact Block	INO + INC
Operation	Break before Make
Nominal rating at 500V, 50 Hz	2A
Terminal capacity: wires upto	2 x 2.5 mm ²
No. of additional block capacity	2
Coil: Voltages available	180V, 220/240V 360/440V, etc.
Consumption: Pick-up: Hold-on:	57 VA 15 VA (15W)
Insulation Voltage:	500 V
Spares Kit Contents:	4 Nos. Each Fix Contacts, front & back, 4 Moving contacts, spring & guides.

DESIGN AND PERFORMANCE

GK-1 contactors are rigid and designed for trouble-free switching of AC motors and other AC loads. They withstand high ambient humidity and frequent starts. Silver-alloy contacts (AgNi / AgCdO) support long life.

CRNGO laminations reduce magnet losses. CNC-wound coils are designed for long life and can withstand wide voltage variations.

TECHNICAL REFERENCE

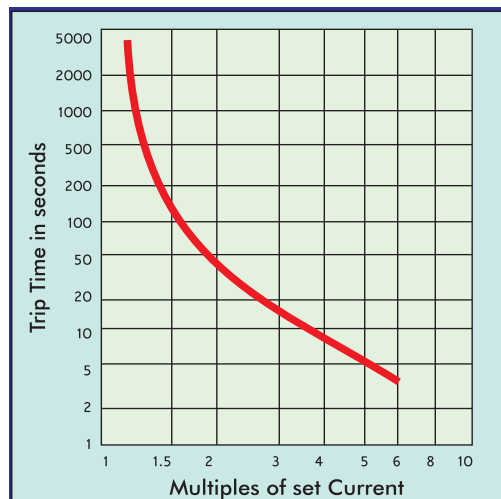
Protection & installation

Overload behavior, mounting pitch and D.O.L. wiring.

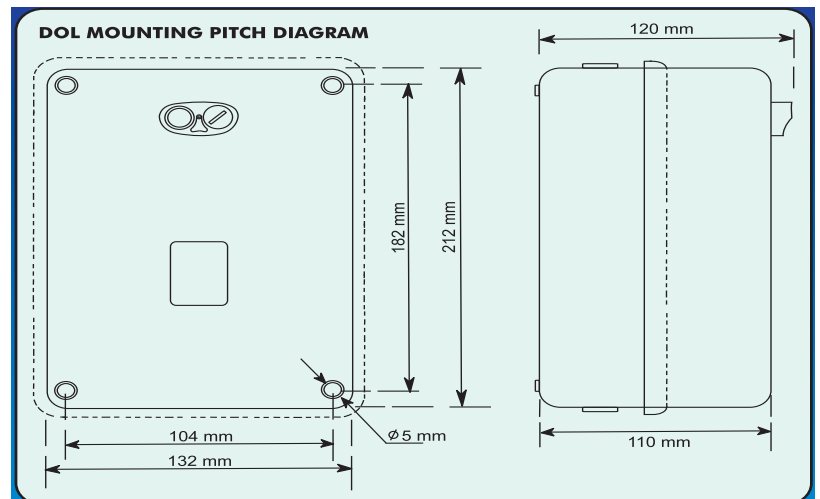
THERMAL OVERLOAD RELAYS

GK-1 relays protect motors against overloads. Their bimetallic elements are individually calibrated and tested. The curve shows inverse-time characteristics from a cold condition. Relay ranges: 0.6-1 A to 20-32 A.

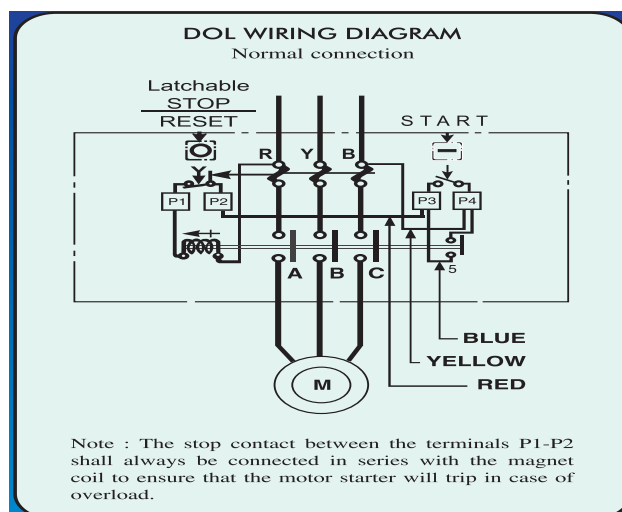
INVERSE-TIME TRIPPING CURVE



D.O.L. MOUNTING PITCH



D.O.L. WIRING DIAGRAM



INSTALLATION REFERENCE

The supplied wiring diagram shows START, STOP / RESET, contactor and overload connections.

Verify coil voltage, motor current, relay setting, and final dimensions for the production KVP unit.